Project Jarvis ??? AI Infrastructure Blueprint v2.0

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Project Jarvis — Personal AI Infrastructure Blueprint

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Executive Summary

This document is a complete implementation guide for building "Jarvis" — a private, voice-enabled AI assistant hosted on a Mac Studio M3 Ultra. The system serves as both a personal AI workstation and a scalable infrastructure for multiple businesses, starting with WealthPilot. It also covers the "Develop Local, Deploy Cloud" production strategy, leveraging India's government AI infrastructure (IndiaAI Mission), and connecting with the Hyderabad/Telangana AI ecosystem.

Core Principles: Data sovereignty, privacy-first design, scalable architecture, voice-first interaction, develop local / deploy cloud.

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1. Hardware Specification

Primary Workstation — Mac Studio M3 Ultra

Component	Specification	Why
Chip	M3 Ultra (28-core CPU, 60-core GPU)	819 GB/s memory bandwidth for fast LLM inference
Memory	256GB Unified	Runs 70B models + dev stack + visualization + headroom for growth
Storage	4TB SSD	Market data, model weights, multiple business data
Display	Dual 4K 27" monitors (minimum)	Screen 1: Code/AI/Jupyter. Screen 2: Dashboards/Charts
Approx. India Price	₹6.5-7L (Mac Studio only)	—

Supporting Hardware

Item	Specification	Approx Cost (INR)
Dual 4K 27" monitors	LG/Dell 27" IPS	₹40-60K

Item	Specification	Approx Cost (INR)
UPS (essential for India)	1KVA pure sine wave	₹15-25K
External NVMe backup	2TB Thunderbolt 5 SSD	₹15-20K
Keyboard + Mouse/Trackpad	Apple Magic or mechanical	₹10-15K
Total Setup Cost	—	**₹7.5-8.5L**

Memory Budget (256GB Allocation)

Component	Memory	Notes
macOS overhead	~10 GB	System processes
Primary LLM (70B Q4)	~40-50 GB	Including KV cache at full context
Secondary LLM (coding 32B)	~20 GB	Qwen2.5-Coder or similar
Whisper large-v3 (STT)	~3-4 GB	Speech-to-text
Kokoro TTS	~1-2 GB	Text-to-speech
WealthPilot stack (Docker)	~20-30 GB	TimescaleDB, paper trading, data pipelines
Visualization stack	~5-10 GB	Grafana, Jupyter, Streamlit
RAG vector database	~5-10 GB	ChromaDB/Qdrant
Future Business #2/#3	~15-20 GB	Reserved headroom
Free headroom	**~60-80 GB**	For larger future models

2. Software Architecture

Layer 1 — AI Inference Engine

Ollama serves as the model server, always running, bound to localhost only.

Models to install on Day 1:

Model	Purpose	Size (Q4)	Command
Llama 3.1 70B	General assistant (Jarvis brain)	~40 GB	`ollama pull llama3.1:70b`
Qwen2.5-Coder 32B	Code specialist	~20 GB	`ollama pull qwen2.5-coder:32b`
Qwen3 72B	Alternative general (test both)	~42 GB	`ollama pull qwen3:72b`

Installation:

```bash
# Install Ollama
curl -fsSL https://ollama.com/install.sh | sh

# Configure Ollama to bind to localhost only
# Edit ~/.ollama/config (or launchd plist on macOS)
export OLLAMA_HOST=127.0.0.1:11434

# Pull models
ollama pull llama3.1:70b
ollama pull qwen2.5-coder:32b
```

Layer 2 — Interface & Orchestration

Open WebUI provides the chat interface with built-in voice support.

```bash
# Docker Compose for Open WebUI
docker run -d \
  -p 3000:8080 \
  -v open-webui:/app/backend/data \
  -e OLLAMA_BASE_URL=http://host.docker.internal:11434 \
  --name open-webui \
  --restart always \
  ghcr.io/open-webui/open-webui:main
```

Access at: `http://localhost:3000`

Voice Configuration in Open WebUI:

1. Navigate to Admin > Settings > Audio
2. STT Engine: Set to "Local Whisper" (downloads and runs Whisper locally)
3. TTS Engine: Set to "Kokoro" or "WebAPI" for browser-native TTS
4. Enable "Hands-free voice mode" for continuous conversation

Layer 3 — Visualization Stack

Docker Compose for the full visualization and data stack:

```yaml
# docker-compose.viz.yml
version: '3.8'

services:
  grafana:
    image: grafana/grafana:latest
    ports:
      - "3001:3000"
    volumes:
      - grafana-data:/var/lib/grafana
    environment:
      - GF_SECURITY_ADMIN_PASSWORD=your_secure_password
    restart: always
```

```yaml
  jupyter:
    image: jupyter/scipy-notebook:latest
    ports:
      - "8888:8888"
    volumes:
      - ./notebooks:/home/jovyan/work
      - ./data:/home/jovyan/data
    environment:
      - JUPYTER_TOKEN=your_secure_token
    restart: always

  timescaledb:
    image: timescale/timescaledb:latest-pg16
    ports:
      - "5432:5432"
    volumes:
      - timescale-data:/var/lib/postgresql/data
    environment:
      - POSTGRES_PASSWORD=your_db_password
      - POSTGRES_DB=wealthpilot
    restart: always

  chromadb:
    image: chromadb/chroma:latest
    ports:
      - "8000:8000"
    volumes:
      - chroma-data:/chroma/chroma
    restart: always

volumes:
  grafana-data:
  timescale-data:
  chroma-data:
```

Launch: `docker compose -f docker-compose.viz.yml up -d`

---

3. Voice System — "Hey Jarvis"

Voice Pipeline Architecture

```
You speak → Whisper (STT) → Text → Ollama (LLM) → Response text →
Kokoro (TTS) → You hear
     ↕                                    ↕
 Microphone                          RAG Context
 (Mac/iPhone)                     (Business docs)
```

At Your Desk (Mac Studio)

Open WebUI provides built-in voice with local Whisper STT and TTS. The experience is: open browser, click mic (or use hotkey), speak, hear response.

For always-listening "Hey Jarvis" wake word, add Home Assistant:

```yaml
# docker-compose.homeassistant.yml
version: '3.8'
services:
  homeassistant:
    image: homeassistant/home-assistant:stable
    ports:
      - "8123:8123"
    volumes:
      - ha-config:/config
    restart: always

volumes:
  ha-config:
```

Configure the Ollama integration in Home Assistant (Settings > Devices & Services > Add > Ollama) and set up the Wyoming voice satellite protocol for wake word detection.

From iPhone (Anywhere in the World)

Setup Steps:

1. Install Tailscale on Mac Studio: `brew install tailscale`
2. Install Tailscale on iPhone from App Store
3. Join both devices to the same Tailnet (login with same account)
4. Enable Tailscale Serve on Mac Studio for HTTPS (required for microphone access):

```bash
# Expose Open WebUI over HTTPS on Tailnet
tailscale serve --bg --https 443 localhost:3000
```

5. On iPhone Safari, navigate to: `https://mac-studio.your-tailnet.ts.net`
6. Add to Home Screen as PWA (tap Share > Add to Home Screen > name it "Jarvis")
7. Open Jarvis PWA, tap mic, speak

Important: HTTPS is mandatory for browser microphone access. Tailscale Serve provides this automatically with valid TLS certificates.

Voice Latency Expectations

Step	Latency	Notes
Speech-to-Text (Whisper)	1-2 seconds	For a typical 5-10 word sentence
LLM Inference (70B, first token)	1-2 seconds	Prompt processing
LLM Generation	15-25 tok/s	M3 Ultra with 70B Q4

Step	Latency	Notes
Text-to-Speech	0.5-1 second	Kokoro is fast
Total round-trip	**3-6 seconds**	Before Jarvis starts speaking

This is usable but not instantaneous. For faster responses, use a smaller model (32B) which roughly halves inference time.

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4. Security Architecture

Network Security

Rule #1: Nothing is ever exposed to the public internet.

All remote access is through Tailscale, a WireGuard-based mesh VPN. Connections are peer-to-peer and encrypted end-to-end.

```
┌──────────────────────────────────────────────────────┐
│          TAILNET (WireGuard encrypted)                │
│                                                       │
│  iPhone ◄──────────► Mac Studio                       │
│  MacBook Air ◄──────► (all services)                  │
│  Future devices ◄────► bound to localhost             │
│                                                       │
│  Public Internet: ZERO open ports                     │
└──────────────────────────────────────────────────────┘
```

Tailscale ACL Configuration:

```json
{
  "acls": [
    {
      "action": "accept",
      "src": ["raj@your-email.com"],
      "dst": ["tag:jarvis:*"]
    }
  ],
  "tagOwners": {
    "tag:jarvis": ["raj@your-email.com"]
  }
}
```

This ensures only your authenticated devices can access the Mac Studio.

Data Security

Layer	Protection
Disk encryption	FileVault (enabled by default on macOS)
Network	Tailscale WireGuard (AES-256

Layer	Protection
	equivalent)
Application auth	Open WebUI login + 2FA
Docker isolation	Separate volumes per business
Model verification	SHA checksum validation on download
Session management	30-minute inactivity timeout

Data Compartmentalization (Multi-Business)

Each business gets isolated Docker volumes and RAG namespaces:

```
/docker-volumes/
├── wealthpilot/
│   ├── timescaledb-data/     # Trading data, portfolio
│   ├── grafana-data/         # WealthPilot dashboards
│   └── rag-collection/       # Strategy docs, research
├── business-2/
│   ├── postgres-data/
│   └── rag-collection/
└── shared/
    ├── ollama-models/        # Shared LLM weights
    ├── open-webui-data/      # Chat history (encrypted)
    └── whisper-models/       # Voice models
```

Voice-Specific Security

Threat	Risk	Mitigation
Voice intercepted in transit	Low	Tailscale WireGuard encryption
iPhone stolen, Jarvis accessed	Medium	Face ID on Tailscale app, Open WebUI session timeout, 2FA
Overheard in public	Medium-High	Use earbuds; switch to text for sensitive queries
Whisper mistranscription	Low	Transcription shown on screen before processing
Tailscale account compromised	Medium	Strong password + 2FA on Tailscale account
Open WebUI zero-day	Medium	Keep updated; Tailnet-only access

Security Hardening Checklist

- ☐ Enable FileVault on Mac Studio
- ☐ Set strong Mac login password + auto-lock at 5 minutes
- ☐ Install and configure Tailscale with ACLs

- ☐ Enable 2FA on Tailscale account
- ☐ Set Open WebUI password + enable 2FA
- ☐ Configure 30-minute session timeout in Open WebUI
- ☐ Bind Ollama to 127.0.0.1 only (OLLAMA_HOST env var)
- ☐ Verify model checksums after download
- ☐ Enable audit logging in Open WebUI
- ☐ Set up automatic macOS updates
- ☐ Configure UPS with auto-shutdown on low battery
- ☐ Create encrypted Time Machine backup to external drive
- ☐ Never expose any port to the public internet

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5. RAG Pipeline (Jarvis Knowledge Base)

RAG (Retrieval Augmented Generation) gives Jarvis access to your business documents.

Architecture

```
Your Documents → Chunking → Embedding Model → ChromaDB (Vector Store)
                                                          ↓
User Query → Embedding → Similarity Search → Top-K Chunks → LLM
Context
```

Setup in Open WebUI

Open WebUI has built-in RAG support:

1. Navigate to Workspace > Knowledge
2. Create collections per business: “WealthPilot”, “Business-2”, etc.
3. Upload documents (PDF, markdown, text, code files)
4. Open WebUI automatically chunks, embeds, and indexes them
5. When chatting, select the relevant knowledge base from the sidebar

Custom RAG for Programmatic Access

For deeper integration (e.g., Jupyter notebooks querying Jarvis):

```python
import chromadb
from chromadb.config import Settings

# Connect to local ChromaDB
client = chromadb.HttpClient(host="localhost", port=8000)

# Create collection for WealthPilot
collection = client.get_or_create_collection(
    name="wealthpilot",
    metadata={"business": "wealthpilot"}
)
```

```python
# Add documents
collection.add(
    documents=["Your strategy document text here..."],
    metadatas=[{"source": "strategy-v2.md", "type": "strategy"}],
    ids=["doc-001"]
)

# Query
results = collection.query(
    query_texts=["What is the current regime detection threshold?"],
    n_results=3
)
```

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6. Workstation Workflow

Daily Workflow Layout

Screen 1 (Primary — Code & AI): - VS Code / Cursor with Ollama-connected AI coding assistant - Terminal with Claude Code for complex sessions (cloud, when needed) - Open WebUI in a browser tab (voice-enabled) - Jupyter Lab for EDA and analysis

Screen 2 (Dashboards & Charts): - Grafana: WealthPilot regime indicators, portfolio metrics, risk dashboard - TradingView (browser): Live stock charts for day trading - Streamlit: Custom interactive analysis tools - Browser: Research, news, documentation

AI-Assisted Analysis Workflow

1. Run feature engineering in Jupyter → generates correlation heatmap
2. Screenshot/save plot → paste into Open WebUI chat
3. Jarvis (with WealthPilot RAG context): "The VIX-momentum cross-correlation suggests regime transition. Run expanding window test to check for look-ahead bias."
4. Execute in Jupyter → iterate with Jarvis

Hybrid Cloud Strategy

Task	Where	Reason
Daily assistant, email, Q&A	Local (Jarvis)	Privacy, always available
Sensitive trading/portfolio data	Local (Jarvis)	Data sovereignty
General coding, business docs	Local (Jarvis)	Good enough quality
Complex architecture decisions	Cloud (Claude/GPT)	Frontier reasoning
Multi-file refactoring	Cloud (Claude Code)	Better for complex code
Regulatory research	Cloud	Better accuracy + web access

7. Scaling Path

Adding Business #2, #3...

Each new business simply needs: 1. New Docker Compose stack with isolated volumes 2. New RAG collection in Open WebUI (or ChromaDB) 3. New Grafana dashboard (if data-heavy) 4. Same AI assistant serves all businesses with context switching

When You Outgrow One Mac Studio

Option A: Add a second Mac Studio/Mac Mini to the Tailnet as a compute node. Load-balance with Ollama's API.

Option B: Dedicated Linux server with NVIDIA GPU for heavy inference. Mac Studio becomes the interface/orchestration layer.

Option C: As open-source models improve (they're closing the gap rapidly), the same 256GB machine runs better models every 6-12 months without hardware changes.

Model Upgrade Path

Timeline	Expected Capability
Now (Feb 2026)	70B models at ~80% of frontier quality
Late 2026	Open-source models likely at ~85-90% of frontier
2027-2028	256GB may run models matching today's frontier

8. Day-One Setup Checklist

Phase 1 — Hardware Setup (Day 1)

- ☐ Unbox Mac Studio, connect monitors, keyboard, mouse
- ☐ Complete macOS setup, enable FileVault
- ☐ Set strong login password, configure auto-lock
- ☐ Connect to home network (Ethernet preferred over Wi-Fi)
- ☐ Install UPS, configure auto-shutdown settings
- ☐ Connect external backup drive, configure encrypted Time Machine

Phase 2 — Core Software (Day 1-2)

- ☐ Install Homebrew: `/bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"`
- ☐ Install Docker Desktop for Mac
- ☐ Install Tailscale: `brew install tailscale` → login → tag as "jarvis"
- ☐ Install Ollama: `brew install ollama`

- ☐ Configure OLLAMA_HOST=127.0.0.1:11434
- ☐ Pull primary model: `ollama pull llama3.1:70b` (~40GB download)
- ☐ Pull coding model: `ollama pull qwen2.5-coder:32b`
- ☐ Deploy Open WebUI (Docker command from Section 2)
- ☐ Configure Open WebUI: create admin account, set 2FA, configure voice

Phase 3 — Visualization & Data (Day 2-3)

- ☐ Deploy visualization stack (docker-compose.viz.yml from Section 2)
- ☐ Configure Grafana: add TimescaleDB as data source
- ☐ Set up Jupyter: install Python packages (pandas, plotly, seaborn, etc.)
- ☐ Deploy ChromaDB for RAG

Phase 4 — Voice & Remote Access (Day 3-4)

- ☐ Configure Whisper in Open WebUI (Admin > Settings > Audio)
- ☐ Configure TTS in Open WebUI
- ☐ Test voice at desk — full conversation loop
- ☐ Enable Tailscale Serve: `tailscale serve --bg --https 443 localhost:3000`
- ☐ Install Tailscale on iPhone
- ☐ Test iPhone access via Tailnet URL
- ☐ Add Jarvis PWA to iPhone home screen
- ☐ Test voice from iPhone

Phase 5 — WealthPilot Integration (Day 4-7)

- ☐ Migrate WealthPilot Docker stack to Mac Studio
- ☐ Configure TimescaleDB with market data
- ☐ Set up Grafana dashboards for regime indicators
- ☐ Create WealthPilot RAG collection with strategy docs
- ☐ Test Jarvis with WealthPilot context queries

Phase 6 — Security Hardening (Day 7)

- ☐ Complete all items in Security Hardening Checklist (Section 4)
- ☐ Review Tailscale ACLs
- ☐ Test access from unauthorized device (should fail)
- ☐ Verify all services bound to localhost
- ☐ Run `nmap localhost` to confirm no unexpected open ports

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9. Ongoing Costs

Development Phase (Current — Mac Studio Only)

Item	Monthly Cost (INR)
Electricity (~80W × 24/7)	₹500-700

Item	Monthly Cost (INR)
Tailscale (free tier, 3 devices)	₹0
Cloud AI (Claude Pro for complex tasks)	₹1,700 ($20)
Total monthly	**~₹2,200-2,400**

Production Phase (After PMS Launch — Add Cloud)

Item	Monthly Cost (INR)
Mac Studio electricity	₹500-700
Cloud AI (Claude Pro)	₹1,700
E2E Networks — Trading engine (c-type VM)	₹8,000-12,000
E2E Networks — TimescaleDB (dedicated)	₹10,000-15,000
E2E Networks — Monitoring/backup	₹3,000-5,000
Static IP (SEBI compliance)	₹300-500
Total monthly	**~₹24,000-35,000**

Compare to: AWS Mumbai equivalent would be ₹30,000-45,000/month (USD billing, forex risk). IndiaAI portal is NOT suitable for always-on production trading (project-based GPU compute only).

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10. Local AI vs Frontier Models — Honest Assessment

Quality Comparison (Feb 2026)

Task	Local 70B Quality	Frontier (Claude/GPT)	Gap
General Q&A	Very Good	Excellent	Small
Email/doc drafting	Good	Very Good	Small
Data analysis discussion	Good	Excellent	Medium
Complex coding	Fair-Good	Excellent	Significant
Financial reasoning	Fair-Good	Very Good	Medium
Creative writing	Good	Excellent	Medium

Practical Guidance

Local Jarvis handles ~80% of daily tasks at sufficient quality. For the remaining 20% (complex architecture, deep debugging, frontier reasoning), use cloud APIs selectively. The hybrid approach gives you both privacy AND capability.

The Gap Is Closing

Open-source models are improving rapidly. The quality gap that was 15-20 points in 2024 has shrunk to 5-10 points. By late 2026, local models on your 256GB machine may match today's frontier quality.

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11. Production Deployment — "Develop Local, Deploy Cloud"

The Principle

Your Mac Studio is your **research lab and command center** — where you think, experiment, and build. It is NOT a production server. Production workloads that require uptime, regulatory compliance, and low-latency exchange access go to cloud.

Architecture

```
┌──────────────────────────────────┐
┌──────────────────────────────────┐
│     MAC STUDIO (Your Desk)       │     │   CLOUD (E2E / AWS Mumbai)
│                                  │     │
│  ✅ Development environment      │────►│  ✅ Live trading engine
│  ✅ Backtesting & research       │ Git │  ✅ Production database
│  ✅ Paper trading                │push │  ✅ Order execution
│  ✅ Strategy prototyping         │     │  ✅ 99.95% uptime SLA
│  ✅ Jarvis (AI assistant)        │     │  ✅ Monitoring & alerts
│  ✅ EDA, feature engineering     │     │  ✅ Audit trail for SEBI
│  ✅ Grafana dev dashboards       │     │  ✅ Automated backups
│                                  │     │
│  Your alpha research stays HERE  │     │  Only execution logic goes
│  Never touches the cloud         │     │  here — no strategy secrets
```

```
└──────────────────────────────────────────┘
└──────────────────────────────────────┘
```

Why Not Host WealthPilot Production Locally?

Concern	Home Mac Studio	Cloud (E2E/AWS Mumbai)
Uptime	Depends on home power/internet	99.95% SLA, redundant infra
Latency to NSE	Hyderabad → Mumbai (~15ms)	Mumbai data center (~1-3ms)
SEBI compliance	Hard to audit "runs on my Mac"	Documented, auditable infra
Power outage	UPS gives 15-30 min	Redundant power, generators
macOS forced update	Trading engine stops	Linux VMs, controlled updates
Resource contention	Jarvis LLM competes with trading	Isolated, dedicated resources
Investor confidence	Weak	Professional, institutional-grade

Phase-Dependent Strategy

Phase 1 (Now): Development & Paper Trading → Host everything on Mac Studio. Ideal for fast iteration.

Phase 2: Live Trading (Own Capital) → Move execution engine to E2E Networks Mumbai. Keep research local.

Phase 3: PMS Launch → Full cloud production with SLAs, monitoring, disaster recovery, audit trails.

Recommended Cloud Provider for WealthPilot

Primary: E2E Networks (Indian provider) - INR billing, no forex risk - Mumbai data centers, low latency to NSE - UPI/NEFT payment support - Indian data sovereignty compliance (DPDP Act, RBI guidelines) - GPU instances available if needed for ML inference - Self-service, spin up in 30 seconds, no quota approvals - Up to 60% cheaper than AWS/Azure

Alternative: AWS Mumbai (ap-south-1) - Larger ecosystem (more managed services) - Better if you need specific AWS services (SQS, Lambda, etc.) - USD billing — forex risk - Hyderabad region also available for redundancy

SEBI Algo Trading Compliance (2025 Rules)

- **Static IP mandatory** for all algo/API trading — register with broker
- One primary + one backup static IP allowed per application
- Static IP can only be changed once per week
- Max 10 orders per second via broker APIs

- Cloud VPS with Elastic/Static IP satisfies this requirement
- **Action:** Get static IP from E2E Networks Mumbai, whitelist with broker

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12. IndiaAI Mission & Government Compute

Overview

The IndiaAI Mission (₹10,300 crore budget) has deployed 38,000+ GPUs including NVIDIA H100, H200, and Google Trillium TPUs. The IndiaAI Compute Portal offers these at dramatically subsidized rates.

Pricing Comparison — GPU Compute

Provider	H100 Rate (₹/hr)	Subsidy	Billing	Best For
IndiaAI Portal	**₹65-116**	Up to 40%	INR	ML training, research
E2E Networks (spot)	₹70	None	INR	Flexible ML workloads
E2E Networks (on-demand)	₹249	None	INR	Production inference
AceCloud	~₹200-250	None	INR	Training, fine-tuning
AWS	~₹330+	None	USD	Enterprise, global scale
Azure	~₹590	None	USD	Enterprise, MS ecosystem
GCP	~₹350+	None	USD	TPU access, ML tools

IndiaAI Portal — Registration & Access

Eligibility: DPIIT-registered startups, MSMEs, researchers, students, government entities.

Registration Process: 1. Register on IndiaAI Compute Portal using Meri Pehchaan (DigiLocker/Parichay/ePramaan) 2. Fill registration form, upload eligibility documents (DPIIT certificate, company registration) 3. Verification by designated officials 4. Submit project proposal: technical approach, intended impact, estimated GPU hours (Bill of Materials) 5. Requests under 5,000 GPU hours: auto-approved 6. Requests over 5,000 GPU hours: PMEC committee review (submit by 25th, approved by 10th next month) 7. Approvals valid for 30 calendar days — must start using within this period

Subsidy: Up to 40% of compute costs. Projects of national importance get priority.

Empaneled Providers: Yotta, E2E Networks, Tata Communications, and others.

What IndiaAI IS Good For (Your Use Case)

- Training regime detection ML models at scale
- Hyperparameter sweeps across strategy variants
- Fine-tuning LLMs on financial domain data
- Experimenting with deep learning signal generation
- Any future AI-centric product development

What IndiaAI is NOT Good For

- Always-on production trading infrastructure (project-based, 30-day approval windows)
- Persistent VM hosting (it's GPU compute, not general cloud)
- Low-latency order execution (not designed for trading workloads)

IndiaAI Mission 2.0 (Announced Feb 2026)

The second phase shifts from infrastructure-building to adoption, with: - UPI-style AI platform offering ready-to-use AI solutions for MSMEs - Expanded sovereign AI model development - Deeper integration with startup ecosystem - Continued GPU capacity expansion (projected 100,000+ GPUs by end of 2026)

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13. Telangana & Hyderabad AI Ecosystem

Why This Matters (You're In Hyderabad)

Hyderabad is rapidly emerging as India's AI capital. Stanford's 2025 Global AI Vibrancy Tool ranks India third globally in AI competitiveness. Telangana is at the center of this push.

Key Resources Available to You

T-Hub (World's Largest Startup Incubator) - 572,000+ sq ft facility in Raidurg, Hyderabad - Supported 2,000+ startups, collectively raising over $2 billion - Programs: Lab32 incubation, acceleration, mentorship, investor access - T-Fund: Early-stage funding from Telangana IT department (₹15 Cr allocated) - Selected under Startup India Seed Fund Scheme (₹5 Cr for eligible startups)

Telangana AI Innovation Hub (TAIH) - Consolidates all state AI initiatives under one umbrella - Connected to 270+ AI startups through Centres of Excellence - Partners: Google, Microsoft, Amazon, NVIDIA, IIIT-H, IIT-H, ISB, BITS - Targets: 50,000 youth skill training, 10,000 jobs in 3 years

Telangana Data Exchange (TGDeX) — Launched July 2025 - India's first state-led digital public infrastructure for AI - Curated datasets and subsidized GPU access - Partners: JICA, IISc Bengaluru, Boston Consulting Group - Founders can fine-tune models without paying premium cloud fees

AI City Hyderabad — 200 acres in Future City - Purpose-built for AI product companies, startups, GCCs, R&D - Yotta building 25,000 GPU AI Supercomputer (4,000 GPUs Phase 1) - Pay-as-you-use access for startups - NTT DATA + Neysa: 400MW cluster with ~25,000 GPUs

Google AI Accelerator — Launched Feb 2025 - Supporting Hyderabad startups in agriculture, mobility, education, sustainability

Startup Telangana Benefits

- Up to 30% reimbursement on international marketing expenses
- SGST reimbursement for eligible startups
- T-Spark grants for early-stage and seed-stage startups
- Access to 500+ Global Capability Centres in Hyderabad

How to Leverage (Action Items)

1. **Register as DPIIT startup** → Unlocks IndiaAI, T-Hub, Startup India benefits
2. **Apply to T-Hub Lab32** → Incubation, mentorship, investor access
3. **Register on TGDeX** → Free/subsidized datasets and compute
4. **Register on IndiaAI Compute Portal** → ₹65/hr GPU access
5. **Track Yotta AI City timeline** → Local GPU cloud when operational
6. **Attend TAIH events** → Networking with 270+ AI startups, potential partners

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14. Complete Infrastructure Cost Summary

One-Time Costs

Item	Cost (INR)
Mac Studio M3 Ultra 256GB, 4TB	₹6,50,000-7,00,000
Dual 4K monitors	₹40,000-60,000
UPS (1KVA)	₹15,000-25,000
External NVMe backup	₹15,000-20,000
Keyboard + Mouse	₹10,000-15,000
Total hardware	**₹7,30,000-8,20,000**

Monthly Costs by Phase

Phase	Local	Cloud	Total/Month
Development (now)	₹2,200	₹0	**₹2,200**
Paper trading + research	₹2,200	₹0	**₹2,200**
Live trading (own capital)	₹2,200	₹22,000	**₹24,000**
PMS launch	₹2,200	₹35,000	**₹37,000**
+ IndiaAI GPU (occasional)	—	₹2,000-5,000	**+₹2-5K**

ROI Perspective

- Hardware pays for itself vs cloud-only in ~2-3 years
- IndiaAI GPU access saves ₹50K-2L/year vs commercial GPU rates
- Data sovereignty and privacy: priceless for proprietary trading strategies
- Telangana ecosystem access: potential funding, mentorship, partnerships at zero cost

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This is a living document (v2.0). Update as you add businesses, upgrade models, evolve architecture, or move through development phases.